



Is the ground really that rocky?

Method, campaign evidence, literature and a test programme for the shadow
module, run 20260912T072640Z

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1 What this document is for

On 12 September 2026, run 20260912T072640Z produced terrain, shadow and fused maps from imagery of the IM-2 Athena landing site. The terrain index was below its configured limit at touchdown. The shadow ranker exceeded its score threshold in 89.1 per cent of assessed cells, and footprint buffering flagged every available shadow-map pixel.

Saturation is not, by itself, proof that the detections are false. A dense population of real hazards could produce this result. On the other hand, systematic model error or shadows from resolved terrain poorly represented by the DEM could produce similar output. The present threshold cannot distinguish safer regions within the available area, regardless of which explanation proves to be correct.

This document examines why this output may have occurred. It explains what the ranker measures, checks the numerical evidence and proposes tests that can discriminate between explanations. The occupancy calculation in section 7 is an illustrative population model, not an inference that the measured threshold-exceedance fraction equals rock occupancy. Its result depends strongly on diameter, height-to-diameter relation, size distribution and spatial clustering.

Mons Mouton's ancient geology makes local rock abundance a relevant question. It does not establish the abundance of exposed sub-metre rocks at this specific location. Geological model ages describe a different quantity from the production, exposure and breakdown history of individual surface rocks.

EDITION 2, 24 SEPTEMBER 2026

This edition keeps the corrected text of 19 September and adds three things. Section 8 summarises published work on lunar rocks, small craters and surface texture; every claim used was checked at its source. Section 11 reports the controls and ablations of campaign athena-watch-01, as documented in the architecture and implementation review of 24 September. Section 13 records the status of each test and adds five new ones.

In that campaign a changing background with no caster produced a warning in all 56 assessable trials, and removing frames or reassigning Sun geometry left most real warnings in place. The literature finds small degraded craters and textured regolith common on old highland terrain, and exposed metre-scale rocks sparse wherever they have been measured. Taken together, real relief broader than the caster template, combined with an uncalibrated noise scale, is now a stronger candidate explanation than either a pure artefact or a dense field of discrete sub-metre rocks. Neither source settles the question for this site.

HOW TO READ THIS

Sections 3 and 4 describe the method. Section 5 separates the original run from the 2.5.5 replay. Sections 6 to 11 examine possible explanations: section 8 covers the published literature and section 11 the campaign results. Section 12 discusses a vehicle-relevant hazard product, and sections 13 to 15 set out tests and decision criteria. Read sections 6, 8, 11 and 13 first if time is limited.

2 The question in one page

A rock's height, shape and position relative to a vehicle's footpads matter for landing stability. However, there is no universal half-metre tip-over threshold that applies to every lander. In these data the projected image posting is 0.9 metres, so some relevant features may be smaller than a pixel; whether their brightness signal is detectable also depends on contrast, optics, noise and geometry.

HATI exploits the shadow rather than relying only on the caster's directly imaged area. On a level receiving surface, a vertical object under a Sun three degrees above the horizon casts a shadow about nineteen times its height. Images with different Sun directions provide a way to distinguish moving illumination structure from static reflectance.

The pipeline estimates subpixel image shifts, but interpolation precision is not registration accuracy. The retained frames have a median global re-correlation residual of 0.0605 pixels; that check does not establish comparable local or absolute accuracy. The stricter local diagnostic in 2.5.5 has very limited supported coverage.

At the fixed touchdown location, the native DEM fit on a 16 metre baseline gives a slope of 2.96 degrees and an RMS height residual of about eight centimetres. The RMS is not the maximum relief: positive and negative residual extremes are approximately 0.187 and 0.113 metres. Terrain warnings cover 1.52 per cent of the available terrain map under illustrative vehicle limits.

The shadow module's output is harder to interpret. A score threshold is exceeded in 89.1 per cent of assessed cells; this is not a measured fraction containing rocks. After buffering, all available shadow pixels exceed threshold. The touchdown's raw cell itself is unassessed, and its displayed warning is inherited from nearby evidence.

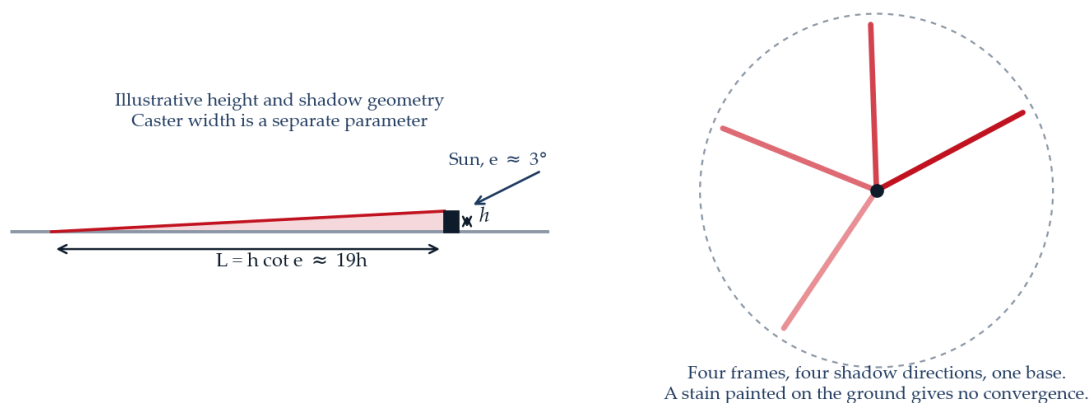
Possible causes include actual unresolved casters, real shadows from resolved terrain absent or distorted in the DEM, photometric differences, and registration or calibration errors. Several may coexist, so the physical explanation should remain open. Saturation alone does not justify choosing the 'unresolved casters' explanation.

3 How HATI measures a shadow

3.1 The shadow law

On a level receiving surface, a vertical object of height h under a Sun at elevation e casts a shadow of length $L = h \cot e$ along the anti-solar direction. The eight retained frames have measured elevations from 3.2755 to 4.3469 degrees, so $\cot e$ ranges from 13.16 to 17.47. A 0.3 metre height therefore gives a level-ground shadow about 3.95 to 5.24 metres long, or 4.4 to 5.8 image pixels. Height specifies neither the caster's other dimensions nor its fractional pixel area. The production renderer also accounts for the receiving slope from module 1 and assumes a finite solar disc.

The shadow gets longer, not wider. That matters twice: it is why a sub-pixel object leaves a multi-pixel trace, and it is why the trace is often a thin streak that a careless cleanup step will delete. HATI's early detector versions did exactly that and found nothing under three pixels wide until the injection tests caught it.



The shadow law in side view, and the sweep in plan. Shadows from a fixed caster rotate about its base as the Sun moves. This is an idealised geometry demonstration.

3.2 Why a sweep and not a single image

A single image cannot tell a shadow from a dark rock face, a dust stain, or a shallow depression. All of them are dark pixels. What separates them is behaviour under a moving Sun. A shadow's far end sweeps around its caster's base at the rate the Sun moves, and its length changes as $\cot e$. Other albedo stains stay put.

HATI therefore collects every usable frame of one site across the range of Sun azimuths available from the LROC NAC, aligns them and analyses the stack. A shadow is distinguished from other albedo features by its movement under the changing Sun. The method has no underlying single-image detector; the sweep itself provides the distinction.

3.3 What the likelihood ranker actually computes

A precise distinction between a template-fit score and an object detection is essential to interpreting saturation.

The shadow module does not threshold dark pixels. For every four-pixel analysis cell it hypothesises a caster at that position with a height from the bank $\{0.3, 0.6, 1.2\}$ metres and a width from $\{0.6, 1.2\}$

metres. For each hypothesis it renders the shadow each frame would see, given that frame's measured Sun azimuth and elevation and the local receiving slope from the elevation model. It then fits one contrast across all eligible frames jointly and asks how much the fit to the observed intensities improves over a null model that has no caster at all. The score is the square root of that improvement in chi-squared. The regional index is $\text{score} / (\text{score} + 8)$. A cell is "flagged" when its score is at least 8.

Each cell jointly fits the frames that satisfy its visibility and data-support requirements; not every retained frame is eligible in every cell. The search covers sixteen sampled root positions per full cell and six height/width combinations per root. A score of at least 8 means a maximum improvement of at least 64 in the model's whitened squared-residual objective. The search maximum is neither a calibrated eight-sigma detection nor a probability of rock occupancy. Frame contributions can be negative, and the nuisance projection couples frames.

3.4 The nuisance model, and why it matters here

A likelihood ratio is only as honest as its null. HATI's null says the observed intensities are a static albedo field, plus a per-frame brightness plane to absorb calibration and phase-angle differences, plus noise with a covariance that accounts for small Gaussian registration jitter of the assumed sigma. Both the data and the shadow templates pass through the same nuisance operator, so the null cannot be gamed by the template absorbing something the null already removes.

What the null does not contain is anything that varies from frame to frame with spatial structure finer than a plane: local registration error of a pixel or more, photometric response differences between the two camera channels, echo-correction residue, or terrain-induced brightness variation that the elevation model is too coarse to predict. If any of those are present, a wide directional template can improve the fit against them almost anywhere, and the ranker may exceed threshold without identifying a discrete unresolved caster.

That is the mechanism behind hypothesis H1 in section 6. Whether it is operating here is exactly what the evidence in sections 9 to 11 argues about.

4 The measurement chain

The measurement chain has 5 stages. Stages one to three are the ground segment and turn an archive of raw detector dumps into a stack of calibrated, map-projected, co-registered images of one patch of ground. Stage four holds the physics. Stage five combines the products without averaging a strong hazard away.

4.1 Acquire

The Orbital Data Explorer is queried for every Narrow Angle Camera observation whose footprint touches the site. The archive filters on a bounding box, and near the pole a long diagonal strip has a box far larger than the strip, so most returned observations do not image the site. The September query returned 696 observations, 637 with footprints. A polygon test in projected polar-stereographic coordinates, returning a signed margin in metres to the strip edge, found that 513 miss the site outright and 64 only clip its edge. Sixty remained with the required 600 metre clearance.

The archive serves no Sun azimuth field. A sub-solar longitude model calibrated against the ISIS camera model gives it to 0.08 degrees over two years. Frames are then binned by azimuth into 24 sectors with ranked alternates per bin, capped at 40 degrees emission angle because relief parallax scales as height times $\tan(\text{emission})$, and the selector reports the azimuth spread as 360 degrees minus the largest gap.

4.2 Ingest

USGS ISIS performs the calibration: `Ironac2isis` to convert the archive product to a cube, `spiceinit` to attach the kernels that say where the spacecraft and Sun were, `Ironaccal` for radiometric calibration, `Ironacecho` to remove the detector echo, and `cam2map` to project into a shared 5 kilometre box at 0.9 metres per pixel.

Between `spiceinit` and the expensive steps sits a gate. The footprint describes the observation, so `camppt` asks the camera model directly where the site falls on the detector. The answer is bounded against the real detector width, 5064 samples or 2532 in summed mode, and the correlation window's worth of pixels is required on either side. When the site falls off the downloaded optic the sibling channel is fetched instead. Cached products carry a processing version and are re-checked for data at the touchdown before they are trusted.

4.3 Register

Projected cubes are aligned to a reference ortho using a large correlation window, mean subtraction and phase correlation. The science run then assesses a central 512-pixel window. An estimated translation can include pointing and map-registration effects. Re-correlating the shifted images gives a median residual of 0.0605 pixels for the eight retained frames.

Re-correlation is partly a self-consistency check. Pairwise closure compares a direct inter-frame displacement with the difference of reference-based shifts. It can reveal inconsistencies, but illumination changes and relief can confound it. One additional frame was removed by the earlier closure gate. That decision does not establish the remaining stack's metre-scale local registration accuracy.

An earlier audit found severe closure failures among some widely separated illumination pairs. Large changes in cast shadows can move a correlation peak without representing a real camera displacement. Similar illumination reduces that confound. The 2.5.5 diagnostic consequently reports apparent offsets,

support, peak separation and boundary flags without applying corrections.

4.4 Measure

The scientific tooling contains four relevant physical analyses, although the production three-map path combines the terrain assessment and shadow-root ranker; the separate kinematics voter is not an extra term in that fusion. The shadow kinematics voter detects anti-solar streaks per frame, back-projects each to its caster's base, and accumulates votes across the sweep; it carries a shuffled-azimuth null and injection controls and passes the currently implemented synthetic controls. Those controls do not establish performance on the real polar stack. The shadow-root likelihood ranker is the module described in section 3.3 and is the one that saturated. The terrain plane fit performs exact least squares on circular windows of the native elevation model and reports slope, RMS residual, signed relief extremes and the 95th percentile residual, indexed as ratio over one plus ratio against configured limits. The DEM shadow predictor marches rays along each frame's measured Sun bearing through the elevation model with lunar curvature and a finite solar disc, producing nominal and conservative visibility under an assumed height envelope.

4.5 Fuse

Terrain and shadow stay separate products and combine by conservative maximum, so a strong hazard in one cannot be diluted by a weak reading in the other. A low index enters fusion only when the model had recorded support to see a hazard there. Incomplete low evidence stays unknown and is painted grey, never zero.

4.6 What the audit spine records

The ingestion stages record verdicts, and map runs record configuration, hashes of the listed scientific source files and input manifest, geometry, coverage and limitations. The portable bundle additionally hashes its aligned-array payload. Nine offline suites cover mathematical, synthetic and product-interface checks. Reproducibility also depends on the actual input rasters, software environment and numerical implementation; matching a manifest alone does not prove identical image bytes or bitwise results across platforms.

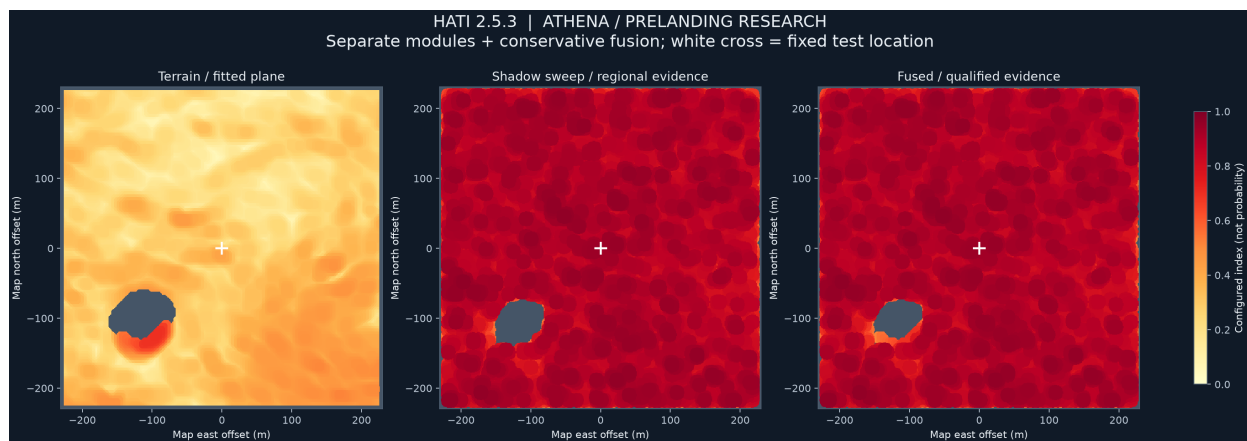
5 What run 20260912T072640Z produced

5.1 The original 2.5.3 run

Quantity	Value	Source
Frames retained	8	run.json
Retained Sun azimuth arc	106.99 degrees	saved map geometry
Median re-correlation residual	0.0605 px	retained coreg_report.csv rows
Analysis window	512 by 512 px; 460.8 m across	run.json
Cells visited / assessed	14 884 / 14 324	run.json
Candidates after deduplication	6 503	candidates.csv
Raw assessed cells above threshold	89.1 per cent	shadow_score.tif
Buffered shadow pixels above threshold	100 per cent of available	shadow_hazard.tif
Terrain pixels above threshold	1.52 per cent of available	terrain_hazard.tif
Terrain / shadow index at touchdown	0.3865 / 0.9377	counterfactual.json
Direct touchdown common support	76.1 per cent; required 80	warning attribution
Controlling root distance	10.58 m; nominal radius 9 m	warning attribution

Values for the original registration-sigma 0.5-pixel scenario. Candidate counts are not independent object counts.

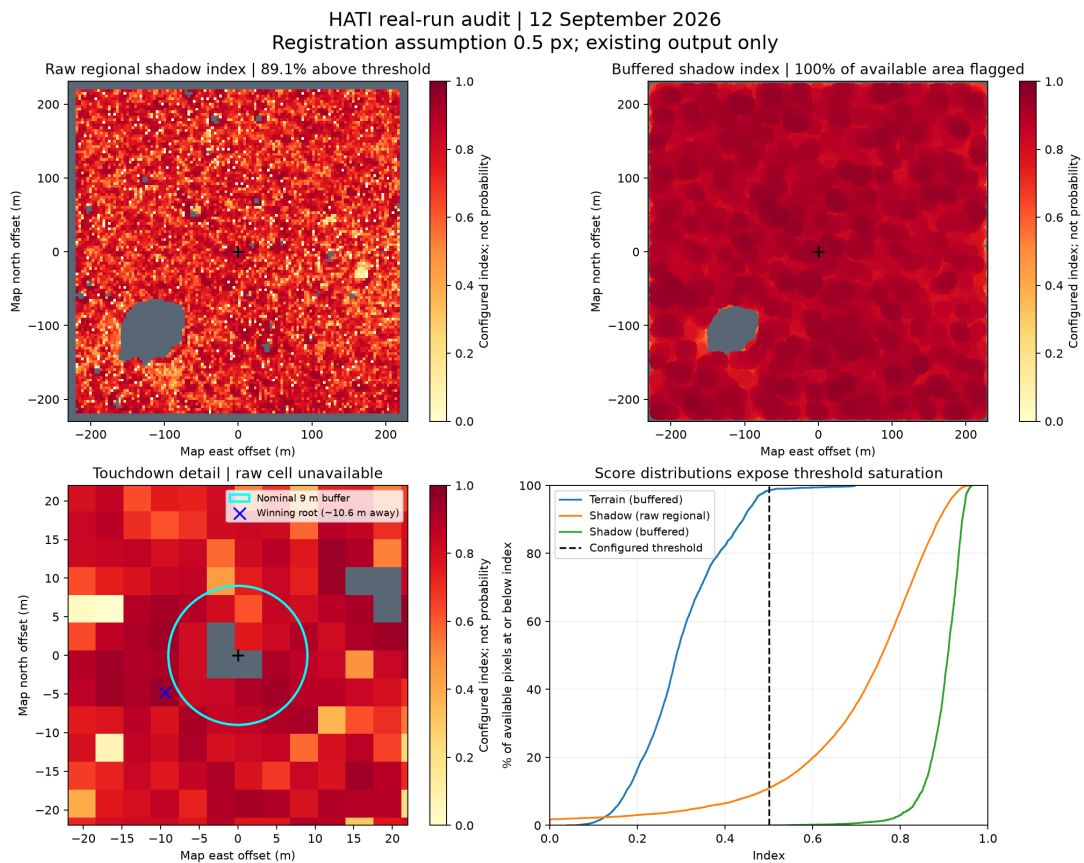
5.2 The three maps



Original 2.5.3 terrain, shadow and fused maps. The cross marks the fixed test coordinate. Grey means unavailable evidence; colour does not establish qualification. All available shadow and fused values exceed threshold.

The terrain index has useful spatial variation under its configured limits. The shadow map also has numerical variation, but its minimum available value is about 0.5007, above the threshold of 0.5. The binary warning therefore does not distinguish the touchdown from other available pixels. This is a limitation of the present decision threshold, not proof that every underlying score is false.

5.3 The saturation audit



Original regional scores, buffered evidence and distributions. Saturation is already substantial before buffering. The threshold-exceedance fraction is not measured rock occupancy.

Raw regional scores have median 24.94 and 95th percentile 81.60, against threshold 8. Footprint buffering increases threshold exceedance from 89.1 per cent of assessed raw cells to 100 per cent of available shadow-map pixels. The two percentages have different support denominators.

5.4 The touchdown warning and the 2.5.5 replay

The original touchdown cell was unassessed because common support was 76.1 per cent, below the 80 per cent floor. Its displayed value came from a nearby cell maximum. Painting that maximum across the cell allowed a root 10.58 metres from the exact coordinate to influence a nominal nine-metre footprint.

Version 2.5.5 retains every sampled root's maximum template score and buffers roots at their actual distances from each output pixel centre. A full cached replay completed without ISIS. Its raw score and terrain rasters are identical to the originals. It retains 229 184 sampled-root scores before deduplication.

In that replay the touchdown shadow index is 0.9292. Its controlling root is 8.82 metres from the exact coordinate and 8.93 metres from the sampled map centre, inside the nominal radius. All available shadow-map pixels still exceed threshold, the raw touchdown cell remains unassessed, and there is still no jointly qualified area. The footprint defect is corrected; the saturation question remains open.

6 Two hypotheses, stated fairly

6.1 H1: the fit is responding to structure outside the intended caster model

H1 includes instrumental or registration residuals, photometric variation and real terrain shadows that the model does not adequately represent. A crater rim can cast a physically consistent shadow without being a subpixel boulder. The latter case must remain distinct from a purely instrumental false alarm: it may still indicate a real landing hazard.

Under H1, the reported threshold-exceedance fraction substantially overstates the evidence for the intended population of unresolved discrete casters. Improving the model or changing the interpretation may be necessary, but reducing warnings is not itself the objective.

6.2 H2: unresolved casters contribute substantially to the widespread evidence

A dense population of small protrusions could make occupancy high on a 3.6 metre grid. This is physically possible in principle. Whether it applies here requires a site-specific size/height distribution and a detector response model, including missed detections, false positives, overlap and selection effects.

6.3 H1 has two parts with different consequences

H1 covers two physically different situations. In one, the fit responds to instrument, calibration or registration residuals, and the evidence says nothing about the ground. In the other, it responds to real relief that the caster template does not describe: small degraded craters and their rims, slope texture, or hummocky ground finer than the DEM. Such relief casts shadows that move with the Sun, so it can raise the score for physical reasons. It may itself be a hazard, but its height, extent and consequence for a vehicle differ from those of a discrete rock. Section 8 shows that the literature expects this kind of relief to be common on old highland terrain. Telling the two parts of H1 apart matters as much as telling H1 from H2.

6.4 Both at once

Instrumental residuals, shadows from resolved terrain and unresolved casters can coexist. Their effects on a maximised fit statistic are not additive occupancy percentages. The test programme therefore seeks evidence that separates these mechanisms rather than forcing one global label onto the entire window.

7 Testing H2 on its own terms

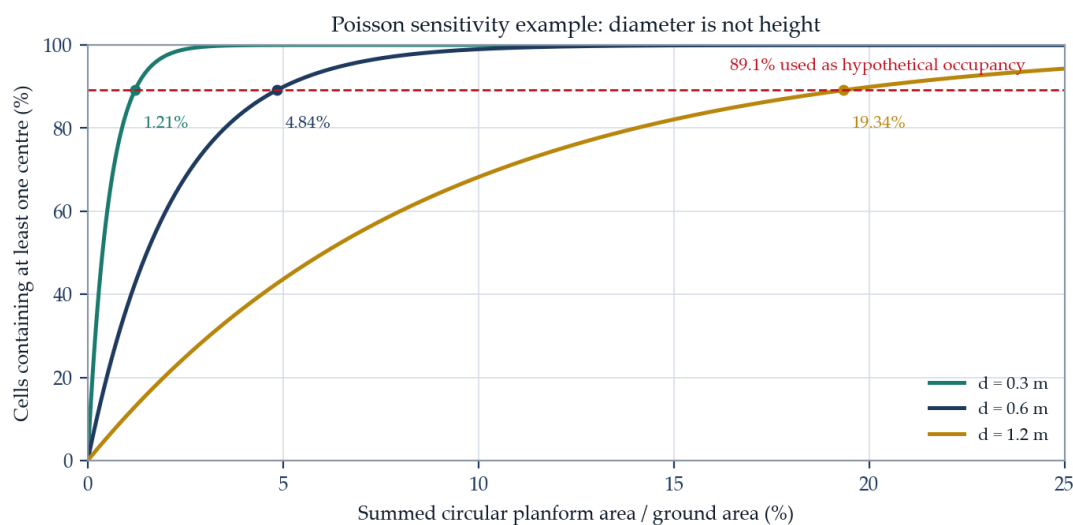
7.1 Occupancy arithmetic

Consider an illustrative homogeneous Poisson population of rock centres. For a cell area $A = 12.96$ square metres and centre density n , the probability of at least one centre is $p = 1 - \exp(-nA)$. Setting p to 0.891 gives $n = 0.1710$ rocks per square metre. This assumes, for illustration only, that occupancy equals the observed threshold-exceedance fraction.

For identical circular planforms of diameter d , summed planform area per unit ground area is $f = n \pi d^2 / 4$. This is approximately fractional coverage when overlap is negligible. The actual union of overlapping planforms and a footprint-intersection probability require different calculations.

Assumed common diameter	Planform area per rock	Summed area fraction giving $p = 0.891$
0.3 m	0.0707 square metres	1.21 per cent
0.6 m	0.2827 square metres	4.84 per cent
1.2 m	1.1310 square metres	19.34 per cent

Sensitivity calculation for identical circular planforms. HATI's bank contains heights 0.3/0.6/1.2 m and widths 0.6/1.2 m; height and diameter are not interchangeable.



Illustrative Poisson occupancy for three fixed diameters. The horizontal line is the observed score-exceedance fraction used as a hypothetical occupancy. No geological plausibility band is asserted.

The earlier 1.2 per cent argument used a 0.3 metre diameter while discussing rocks at least 0.3 metres tall. That conversion is unjustified. Even treating template width as a circular diameter is only a sensitivity example: the bank's width is not a measured rock diameter.

7.2 What geology can contribute

Independent rock-abundance data can constrain a population model. Diviner and NAC images respond to different spatial scales and physical properties, however, and extrapolating from one to the other requires a supported size-frequency model.

The approximately 4.26-billion-year geological model age discussed for this region does not determine modern exposed sub-metre rock abundance. Production, exhumation, burial and breakdown must also be considered. Published work on other lunar terrains demonstrates that surface-rock abundance can decrease with age; it cannot simply be assumed to increase because terrain is old.

Section 8 summarises what published work does and does not constrain, including a conditional occupancy calculation under stated size-distribution assumptions.

7.3 Where this calculation is weak

For a size distribution, $f = n$ times the mean planform area. Height-to-diameter relation, clustering, finite rock extent and overlapping shadows also matter. Most importantly, detector threshold exceedance is not occupancy. The calculation neither establishes H2 nor rules it out, and the size of its uncertainty has not been bounded to a factor of two.

8 What the published literature constrains

A literature search through the Consensus service on 24 September 2026 identified the studies used here. Each claim that bears on the argument was checked against the paper or its abstract, and claims that could not be confirmed at source were left out. This is a targeted search, not a systematic review. The absence of a study here does not show that none exists.

8.1 Rocks: what orbital data measure, and where

Thermal rock abundance from the Diviner radiometer is the main global constraint on exposed rocks. It corresponds to the areal fraction of rocks roughly one to two metres across and larger (Powell et al. 2023). Where the retrieval is valid, rock abundance is low over most of the surface and high mainly around young craters (Bandfield et al. 2011). The original retrieval covers latitudes within 60 degrees of the equator, and later products are reported to extend to about 80 degrees. The Athena site, at 84.8 degrees south, lies outside both, so no thermal rock abundance exists for this window.

Optical counts in NAC images reach rocks a few pixels across, about one to three metres at NAC scales (Li and Wu 2018). Chandrayaan-2 OHRC images, at 0.25 to 0.28 metres per pixel, resolve boulders below one metre. One study mapped more than 2 000 boulders around a young crater at 74.9 degrees south under a six-degree Sun (Dagar et al. 2022).

No dataset used in this project measures rocks between 0.3 and 1 metre at this site. Any statement about that size range depends on an assumed size-frequency distribution. How thermal rock abundance itself depends on the boulder size distribution is examined by Amaro et al. (2026).

8.2 A conditional occupancy calculation

Section 7 showed that 89.1 per cent occupancy by bodies 1.2 metres across would need a summed planform area of 19.3 per cent. That is far above the rock abundance measured over most of the Moon, so a dense population of metre-scale rocks is not a credible explanation for the saturation, even though this latitude has no direct measurement.

Smaller rocks need an assumed size distribution. The table below assumes a power law in which the number of rocks larger than diameter D scales as D to the power minus b , continuing below one metre, with 0.5 per cent areal coverage by rocks at least one metre across. That coverage is chosen for illustration and is not measured here. Rock centres are placed at random, as in section 7, and the mean planform area of the power-law population is used instead of a single diameter.

Exponent b	Cells holding a rock at least 0.3 m across	Cells holding a rock at least 0.6 m across
2.5	28 per cent	6 per cent
3.0	64 per cent	12 per cent
3.5	91 per cent	19 per cent

Illustrative calculation, not a site measurement. At 0.1 per cent coverage by rocks at least one metre across every density falls fivefold, and the 0.3 metre value for $b = 3.5$ becomes 38 per cent.

Under these assumptions, rocks at least 0.3 metres across reach the observed fraction only with the steepest distribution tested and the chosen coverage. Rocks at least 0.6 metres across stay below 20 per cent. Which column applies to the bank's 0.3 metre height depends on the height-to-diameter relation, which is not established here. Exponential distributions, used for Mars landing sites (Golombek and

Rapp 1997) and fitted to lunar counts (Li and Wu 2018), fall off differently and would need parameters fitted for this terrain.

8.3 Extended relief is common on old highland and polar terrain

Cai and Fa (2020) mapped metre-scale roughness in 462 regions from NAC terrain models. Highland roughness exceeds mare roughness by factors of 2.4 in bidirectional slope, 3.0 in RMS height and 2.7 in median absolute slope, and roughness from metre to hectometre scales is mainly controlled by small degraded impact craters. Small craters on lunar surfaces are commonly in equilibrium, with cumulative density falling roughly as the inverse square of size (Xiao and Werner 2015; Minton et al. 2019).

Near the south pole, elephant-hide texture forms lobate patterns decametres to about a hundred metres wide. It is most common on slopes of 5 to 30 degrees but occurs with similar likelihood on shallow slopes of 5 degrees or less, in both illuminated and permanently shadowed terrain (Brown et al. 2025, conference abstract). Its relief amplitude is poorly quantified.

Such relief casts shadows that move with the Sun. At three to four degrees of elevation, a rim or crest a few decimetres high casts a shadow several metres long. The ranker's template describes a compact caster, but it can still improve its fit against these shadows, and section 11 shows it moving toward the widest template on offer. Extended relief is therefore a physically grounded candidate for much of the evidence, and a candidate hazard in its own right.

8.4 Prior art

Shadow-based rock detection from orbit is established on Mars. Golombek et al. (2012) fitted ellipses to shadows and cylinders to rocks in HiRISE images of the Mars Science Laboratory candidate sites. Rocks larger than about 1.5 metres were fully resolved, their size-frequency distributions matched the exponential models derived from lander measurements, and shadows of non-rock features were removed by fitting model distributions. Those rocks span about five HiRISE pixels. HATI targets casters below one NAC pixel, roughly an order of magnitude smaller relative to the pixel, and relies on the sweep to compensate. That compensation has not been demonstrated.

On the Moon, Li and Wu (2018) detected rocks automatically from illumination gradients in NAC and Chang'e-3 descent images, using the fact that rocks and craters show opposite orders of bright and dark along the Sun direction. Cadogan (2020) matched craters across overlapping NAC images taken under different illumination, detected craters down to about three metres, and used agreement between images to reject false positives. The search found no published method that uses a multi-azimuth sweep to detect casters below the pixel scale.

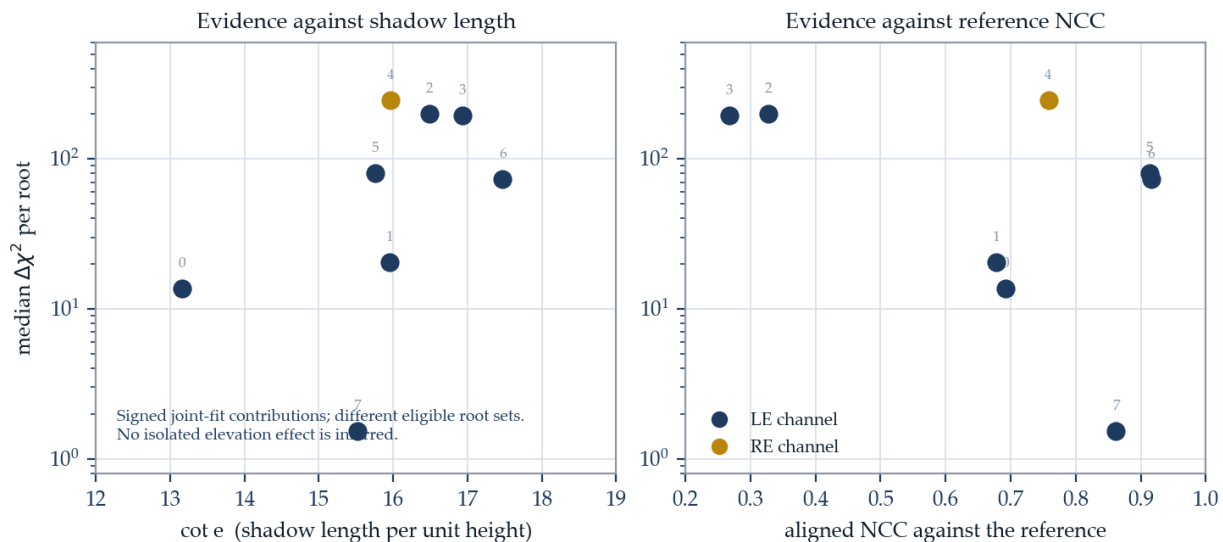
8.5 Where the literature is thin

Three gaps matter for this project. Sub-metre rock abundance has not been measured from orbit at polar latitudes. The relief amplitude of elephant-hide and similar texture is poorly quantified. No lunar study found here validates sub-pixel shadow detections against ground truth under polar low-Sun conditions. Each gap is a reason to seek independent ground truth rather than to infer occupancy from the ranker.

9 Evidence that resists H2

9.1 Frame contributions are strongly uneven

The retained elevations span 3.2755–4.3469 degrees. Corresponding cot e spans 13.16–17.47. Median signed fit contributions vary much more widely, from 1.53 to 247.05, but shadow length alone does not predict those contributions.



Descriptive per-frame median signed fit contributions against the saved measured elevation and aligned NCC. These values are conditional on selected roots and frame eligibility; neither plot isolates a causal effect. Point numbers identify the table rows below.

Frame	Map azimuth	Elevation	cot e	NCC	Median delta	Positive	Roots included
0: m1409532146le	43.13	4.3469	13.16	0.692	13.72	63.5%	6292
1: m1266635336le	43.54	3.5874	15.95	0.678	20.53	67.6%	6279
2: m1169496158le	68.97	3.4711	16.49	0.327	198.51	97.1%	6351
3: m1138847117le	73.66	3.3807	16.93	0.267	194.45	93.6%	6352
4: m1433043518re	326.66	3.5835	15.97	0.758	247.05	85.6%	5587
5: m1379255373le	354.63	3.6319	15.75	0.913	80.22	83.3%	6010
6: m1113828140le	5.97	3.2755	17.47	0.916	73.24	82.8%	6075
7: m124401181le	24.94	3.6870	15.52	0.862	1.53	51.4%	6220

Median delta is the signed change in chi-squared. Geometry comes from the source run's azimuths_map and elevations arrays; NCC from coreg_report.csv. Contributions are matched through each candidate's saved frames list. Denominators vary with frame eligibility. Azimuths are map-grid bearings in degrees, not selection-stage estimates.

The only RE frame and the two lowest-NCC frames contribute strongly. This motivates controlled frame and calibration checks. It does not establish that those frames are defective: illumination direction, receiving slope, support, contrast, occlusion and the joint nuisance projection affect the signed contributions.

9.2 Width accumulates at the bank maximum



Selected template dimensions among deduplicated warning roots. These are fit parameters under selection and censoring, not measured rock dimensions or a recovered size-frequency distribution.

The selected heights are 0.3 metres for 3 577 roots, 0.6 metres for 1 850 and 1.2 metres for 1 076. That decreasing sequence is not evidence of a physical size-frequency distribution without accounting for template sensitivity and selection under realistic null and injected scenes. Noise or structured mismatch can also prefer smaller templates.

Of 6 503 retained roots, 6 052 choose the widest template, 1.2 metres. Bank-edge accumulation warrants a width-profile and model-mismatch test. It does not prove that the signal is nonphysical, because a 0.3-metre-high, 1.2-metre-wide rock is geometrically possible.

9.3 Censoring limits height interpretation

The endpoint-censored flag is true for 77.1 per cent of retained roots. In this implementation it is a geometric diagnostic of the selected template: an endpoint in at least one selected frame extends beyond the root-support region after allowing for root offset and the finite solar disc.

The fitting patch is 25 pixels across, but the common circular root support has radius six pixels, with an additional margin in the censoring check. Root offsets and receiving slope reduce the available length further. Consequently a small caster can be flagged as censored even when its nominal level-ground shadow would fit inside the larger square patch.

A genuinely observed shadow continuing beyond valid support may constrain a lower bound under a valid geometric model. The present censoring flag alone does not provide that bound. Estimated heights require identifiability checks and uncertainty validated under censoring and model mismatch.

10 Evidence that goes against H1

10.1 Synthetic controls demonstrate capability within their tested conditions

The kinematics voter's current synthetic tests reject the planted crater, ridge and static-stain examples while recovering planted casters. The likelihood ranker's favourable small benchmark also recovers small injected casters and does not flag every static scene. Those results justify further investigation. Controls from the voter cannot be transferred to the ranker.

10.2 Stability across registration assumptions

Raw-score Spearman correlations against the 0.25-pixel scenario are approximately 0.9987 and 0.9785 for the other two assumptions, with high spatial candidate correspondence. A persistent model error, shared image noise or a real feature can all remain stable because the scenarios reuse the same observations.

10.3 The translation control

The earlier audit compared actual candidate correspondence with 100 toroidal translations of one candidate field. Actual correspondence exceeds the median shifted correspondence, approximately 64 per cent. This demonstrates spatial structure relative to that specific control. It does not exclude noise-driven saturation, since noise and residual structure are shared between the scenario runs.

10.4 The discrepancy layer tests a narrow failure mode

Version 2.5.5 flags broad dark structure despite nominal DEM illumination in 4.27 per cent of the window, including the touchdown pixel in all eight nominally lit frames. It assesses 97.60 per cent of the window under its support rule.

That layer requires sufficiently dark, broad and repeated components. Photometric gradients, narrow residuals, bright structure and other mismatches may never satisfy those conditions. The remaining area is therefore not a measured clean fraction and cannot be used as evidence against general model mismatch. At touchdown, the flag is consistent with resolved image structure missing or distorted in the DEM; it does not identify its cause.

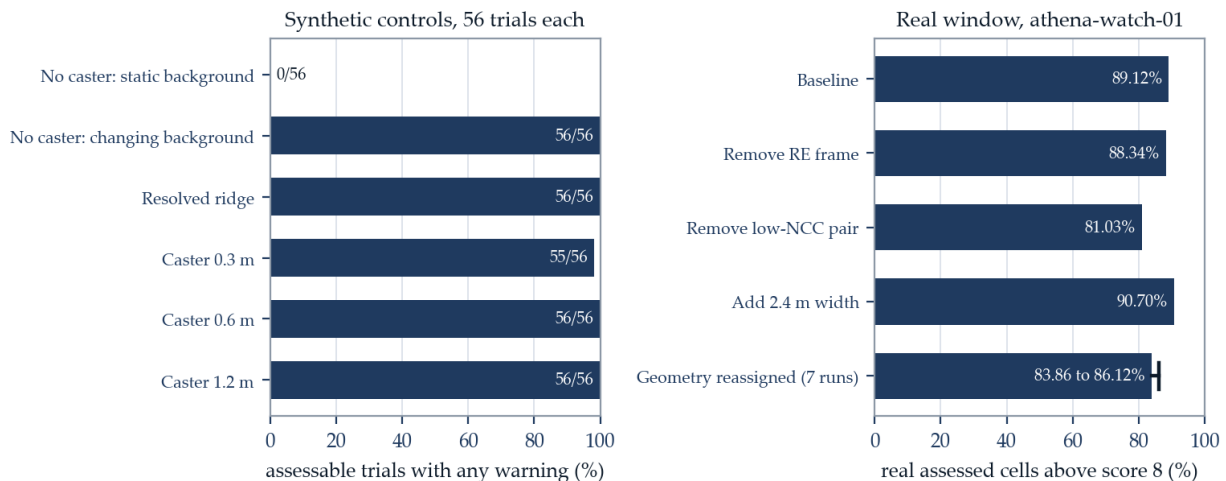
10.5 The albedo-gain experiment

The 2.5.4 fixed-grid diagnostic found warnings in 31 of 61 baseline patches and 32 of 61 gain-model patches. Median null residual energy per degree of freedom decreased from 13.50 to 12.37. The simple gain nuisance did not solve the problem; the residual-energy statistic itself remains conditional on the assumed noise and nuisance model.

The true geometry beat all three cyclic geometry stress scores in 25 of 61 baseline patches and 32 of 61 gain patches. Among training-selected warnings, held-out signed improvement was positive in 18 of 24 and 20 of 26 respectively. These are limited diagnostics with different selected sets, not false-alarm probabilities or detection accuracies. They already supply a small geometry-stress experiment; a full regional version remains to be implemented and validated.

11 Campaign results, athena-watch-01

The architecture and implementation review of 24 September 2026 documents a sequential campaign on the cached Athena products at commit 8347e6d, archived as athena-watch-01. Its maps reproduce the 2.5.5 replay in section 5: 14 324 of 14 884 cells assessed, 89.12 per cent of them above threshold, a touchdown shadow index of 0.9292 and no jointly qualified area. The stages below correspond to tests T2, T3 and T5 to T7 in section 13. They use synthetic controls and the cached real stack; none is independent field validation.



Left: warnings in the synthetic control scenarios, 56 assessable trials each. Right: real assessed cells above score 8 in each ablation, with the range over seven geometry reassignments. Numbers are transcribed from the architecture and implementation review, campaign athena-watch-01.

11.1 Controls: sensitivity without specificity

Each control scenario had 72 planned trials, of which 56 were assessable. The static background without a caster never warned. The changing background contains no rendered caster, only a changing intensity term, texture, a stain and noise, and it warned in all 56 trials. Resolved ridges also warned in all 56, but a ridge is real relief and its warning is not automatically a false hazard. Casters of 0.3, 0.6 and 1.2 metres were localised within two pixels in 55, 56 and 56 trials.

Structured-null maximum scores ranged from 16.32 to 21.52, while the median maximum for the 0.3 metre caster was 17.02. A threshold above every structured-null score would leave a warning in at most 4 of 56 small-caster trials. The two distributions overlap, so raising the threshold does not help. Under these controlled conditions the ranker's small-caster sensitivity comes without the ability to reject a changing background.

11.2 Geometry reassignment

Under seven cyclic reassignments of the measured Sun geometry, the median cell score fell from 24.94 to between 17.93 and 19.54, while 83.86 to 86.12 per cent of assessable cells stayed above threshold. Correct geometry therefore carries information, and the score is not blind to it. Most warnings survive without it, which is consistent with generic changing structure. These reassignments are stress tests, not exchangeable permutation samples.

11.3 Frame ablations and width

Variant	Cells above score 8	Median cell score	Small-caster recovery
Baseline	89.12 per cent	24.94	55 of 56
Remove RE frame	88.34 per cent	21.40	51 of 56
Remove low-NCC pair	81.03 per cent	17.38	48 of 56
Add 2.4 m width	90.70 per cent	27.86	55 of 56

Ablations from athena-watch-01. Removing frames lowers both real scores and matched sensitivity; it is not an independent reason to discard those frames.

No single frame or tested pair carries the saturation. When a 2.4 metre width was added to the bank, 89.09 per cent of deduplicated warning roots moved to the new widest bin. With the accumulation at 1.2 metres described in section 9, this shows the fit preferring structure broader than the bank offered. It is not a measured body width.

11.4 Height and registration

For isolated-caster controls, the imposed height fell inside the descriptive compatible set in 42 of 144 assessable cases at six-pixel support and 13 of 168 at ten-pixel support. Median height errors were minus 0.05, minus 0.075 and minus 0.30 metres for 0.3, 0.6 and 1.2 metre casters. The baseline fitting disk has a radius of six pixels, while the bank's 0.6 and 1.2 metre heights cast level-ground shadows of about 9 to 23 pixels at the measured elevations. Heights above the smallest bank value are therefore mostly censored in the baseline, and their estimates run low.

Planted integer registration shifts were recovered in all 186 measured trials. However, 2 740 of 2 926 tile trials lacked support, and only 30 real tile measurements existed, all with offsets of zero or one pixel. Local registration remains essentially unmeasured.

11.5 The local-plane guard near touchdown

Four cached cells near the fixed coordinate exceeded the adaptive extension's local-plane departure limit, with proxies of about 0.20 to 0.29 metres against a limit of 0.15 metres, and stayed unresolved. A single plane does not describe the receiving surface there at the patch scale, which agrees with the broad-darkness flag at the same location. No full adaptive workstation run had completed at the time of the review.

11.6 The noise scale

The ranker assumes a normalised noise sigma of 0.03 and gives every frame the same noise scale. The 2.5.4 diagnostic reported a median null residual energy of 13.50 per degree of freedom at that sigma, where a null fitting down to the noise would give about one. The whitened residual is therefore about 3.7 times larger than the model expects, from unmodelled structure, underestimated noise, or both.

Scores scale inversely with the noise sigma. If the whole excess were noise, every score would shrink by about 3.7 times, taking the median of 24.94 to about 6.8. If the excess is structure, rescaling is the wrong remedy and a structural term belongs in the null. A shared noise scale may also let the frames with the largest unmodelled residuals dominate the evidence, which is one untested explanation for the uneven frame contributions in section 9. The synthetic controls were rendered with the assumed noise, so their sensitivities may not carry over to the real stack. Test T12 in section 13 is designed to separate these cases.

12 Why the question is badly posed either way

12.1 A score threshold is not a vehicle hazard model

The current exceedance map does not discriminate safer locations within its available area. Even if many signals are real, the vehicle-relevant question involves obstacle height and extent, local slopes and relief, footpad placement, navigation uncertainty and stability margins. It cannot be reduced to whether any template improves the fit.

12.2 Height must supplement evidence and observability

The bank already contains height parameters, but a winning bank value is not a height measurement. A height-based hazard product would need a model with demonstrable identifiability, calibrated uncertainty, treatment of censored or overlapping shadows and verified vehicle tolerances.

If evidence and a validated uncertainty bound establish an obstacle above a vehicle's tolerance, the location can be flagged under that vehicle model. An uncertain estimate or inadequate sensitivity remains unknown. A low fitted height is not clearance: another unresolved hazard may be missed, and incomplete observations cannot justify a safe label. A nominal 68 per cent interval including zero does not by itself establish either safety or danger.

12.3 Engineering priorities

Preserve the separate terrain and shadow maps and their fused product. Develop height profiles and explicit endpoint observability alongside model-adequacy and registration checks. A larger valid fitting region can help with censored shadows, but it may also introduce more overlap, variable slopes or photometric structure. Validate the resulting tradeoff with independent rendered scenes and held-out real data before promoting height estimates into operational thresholds.

The adaptive extension now implements windows of 25, 49 and 97 pixels with fitting radii of 6, 12 and 24 pixels, together with joint height and width refinement. Its full workstation evaluation is pending, and test T16 in section 13 should come before any interpretation of its real-data output.

13 The plan

The following tests are complementary. Their outcomes constrain hypotheses; none has a universal binary interpretation. All thresholds and exclusions should be declared before evaluating held-out data. Start annotation and external-data feasibility work early rather than waiting until the last test.

Test	Campaign stage	Status on 24 September 2026
T1 physical response	none	Open
T2 frame ablation	T2	Run: saturation survives every tested ablation
T3 geometry stress	T3	Run: median score falls by about a quarter; most warnings persist
T4 thermal comparison	T4	Blocked: needs an external thermal product; published retrievals found stop at about 80 degrees
T5 width bank	T5	Run: 89 per cent of warning roots move to the new widest bin
T6 endpoints and height	T6, T9	Partial: heights biased low; adaptive windows implemented, full run pending
T7 local registration	T7	Partial: 30 supported tile measurements
T8 annotated scenes	T8	Implemented for the baseline engine; blocked on inputs

Campaign stage names follow the architecture review, whose T9 to T11 are the adaptive extension, independent mesh controls and withheld-frame prediction. Tests added in this edition are numbered from T12.

13.1 T1. Check physical response using identifiable comparisons

First regenerate geometry and contributions from the saved run, preserving negative per-frame contributions and eligibility masks. Do not regress log evidence on both frame indicators and log cot e : elevation is constant within each frame, so its coefficient is confounded exactly with unrestricted frame effects. Per-root replication does not remove that problem, and negative contributions have no real logarithm.

Use independent forward-rendered scenes to derive the expected score response under the actual geometry, masks, noise, slopes and nuisance projection. Compare controlled geometry/height changes and matched real subsets where possible, allowing for spatial dependence and selection. There is no universal unit slope between log evidence and log cot e . The cost starts with saved-product analysis and grows with the simulation design.

13.2 T2. Frame ablation with matched controls

Run systematic leave-one-frame-out and declared grouped ablations. Keep coordinates fixed and report changes in valid support, Sun arc, receiving-plane geometry, template sensitivity and evidence. Compare like-for-like supported roots and use injections with the same subsets to estimate the loss of signal expected from fewer or less diverse frames.

The RE frame and low-NCC frames are useful diagnostic subsets, not pre-labelled bad observations. A lower score after dropping them does not establish instrumental excess, and saturation in a three-frame complement does not nearly confirm H1. Reprocessing cached inputs needs no ingestion, but exclusions from production require independent quality criteria.

13.3 T3. Geometry stress controls on the regional ranker

Extend the existing three cyclic controls from the small diagnostic bank to the full regional search, using multiple predeclared geometry reassignments. Keep observed masks and preprocessing fixed, document how templates and slopes are handled, and include planted casters, resolved-terrain shadows and structured no-caster scenes.

A geometry-sensitive result supports sensitivity to illumination structure, not uniquely to subpixel boulders. Crater shadows can also follow the Sun. Dense superposed shadows may retain substantial scores under reassignment. Because masks, radiometry and geometry are not freely exchangeable, these are stress controls rather than an automatically calibrated permutation null. The cost is an implementation change and multiple replay runs.

13.4 T4. Independent thermal and geological comparison

Check whether a Diviner rock-abundance product valid for this latitude has enough independent spatial support. Its nominal detector footprint is about 179 by 307 metres at 50 km altitude, so the 461-metre HATI window alone offers limited independent thermal samples. Product gridding can be finer than the physical footprint.

Use a larger surrounding area, product-specific coverage and uncertainty, and aggregate HATI evidence to the effective thermal support. Thermal and optical methods probe different physical scales; correlation can corroborate broad structure but is not a sub-metre object validation or height calibration. Do not infer an arbitrary size-cut conversion from a generic multiplier.

13.5 T5. Expand and profile the width bank

Adding a 2.4-metre width is a useful boundary diagnostic. Track full score profiles, identifiability and performance on known injected features, rather than only the winning bin. The larger search bank also changes the maximum statistic and needs matched null controls.

Migration to the new widest bin indicates unresolved scale or model preference; it does not uniquely identify nonphysical structure. Staying at 1.2 metres does not validate a rock interpretation. This test requires one configuration change and a controlled set of replays.

13.6 T6. Test endpoint observability and height uncertainty

Increase valid support where possible, model shadow endpoints and overlap explicitly, and distinguish an endpoint predicted outside support from an endpoint observed to continue beyond it. Recover known heights with and without censoring using independent rendering and real annotated structures where available.

Report height profiles, unresolved cases, bias and uncertainty coverage. A larger patch alone does not produce an honest height measurement. This is a substantial engineering and validation task.

13.7 T7. Improve local registration coverage without inventing support

The strict 96-pixel tile diagnostic currently yields only one supported tile across two similar-illumination pairs, with a one-pixel apparent offset. It provides no regional error estimate.

Investigate why the shared mask loses support, then evaluate any revised weighting, tile size or missing-data treatment on planted shifts with realistic holes and illumination changes. Do not simply relax the mask until the desired coverage appears. Independent landmarks or stereo control can help distinguish real displacement from shadow movement. Report unsupported and ambiguous areas explicitly.

13.8 T8. Held-out annotated scenes

Identify independent polar scenes and annotation sources early, checking their resolution and completeness. Missing an unannotated subpixel rock is not the same as detecting a false positive. Boulder diameters alone cannot validate heights without a justified geometric relation or independent height information.

Use separate selection, calibration and held-out evaluation sets. Report detection/false-alarm tradeoffs, localisation, size-dependent recovery and abstention under the intended vehicle-relevant thresholds. New imagery, if required, is ingested only on the stationary ISIS machine. No single annotated site by itself establishes a mission-loss bound.

13.9 T12. Measure the noise scale, overall and per frame

Estimate residual variance from the stack itself, per frame and overall, in areas with low scores, nominal DEM illumination and a planar receiving surface. Compare it with the assumed 0.03 and report whether the excess behaves like independent noise or has spatial structure. Then rerun the controls with rendered noise matched to the measured level, and report small-caster sensitivity and changing-background response at that level. This needs only saved products.

13.10 T13. Compete compact, extended and depression models

Add explicit alternatives to the compact caster: a smooth bump and a shallow bowl several metres across, each rendered with its shading and cast shadow under every frame's Sun. Compare the models per cell by withheld-frame prediction, campaign stage T11, rather than by fitted score alone. Include the order of bright and dark along the Sun direction, which is opposite for protrusions and depressions (Li and Wu 2018). Report where each model predicts held-out frames best, and where none does.

13.11 T14. Shape-from-shading as a structural null

The aligned stack, the measured Sun geometry and the coarse DEM are the inputs a multi-image shape-from-shading solver needs. Solving for a surface at an effective resolution of two to four metres, with a smoothness prior, could resolve texture and small craters while leaving sub-metre casters in the residual. Its predicted shading could then enter the null model, and its surface could give the terrain module footprint-scale support. Shape-from-shading terrain models of the Artemis III candidate regions have been published at five metres, and shape-from-shading at native image resolution has been applied at Mons Mouton. The risk is that the solver also absorbs real caster shadows, so injection controls must measure that loss before any adoption.

13.12 T15. Look for OHRC coverage

Check whether Chandrayaan-2 OHRC images at about 0.25 metres per pixel cover the analysis window. OHRC has resolved sub-metre boulders under low Sun at 74.9 degrees south (Dagar et al. 2022), and multi-view OHRC images have been used for sub-metre terrain models in a 2026 preprint. Coverage of this window would give direct boulder counts, and possibly footprint-scale topography, for comparison with HATI's evidence. A post-landing image would need the area disturbed by the landing excluded.

13.13 T16. Run structured nulls through the adaptive procedure

The adaptive extension searches a larger bank with refinement at 0.1 metres and keeps the score threshold of 8. A larger set of hypotheses raises the maximum score expected under any null. Run the

static and changing-background controls through the complete adaptive request, expansion and stopping procedure, and report how often they reach context-supported status, before interpreting any adaptive result on real data.

14 A decision tree

Read the tests as evidence updates rather than a classifier that chooses H1 or H2 from a few thresholds.

Finding	Supported interpretation	Next check
Correct geometry consistently improves held-out fit beyond matched controls	Some illumination-dependent structure is being modelled	Distinguish resolved terrain shadows from unresolved casters
Removing a frame changes evidence beyond matched support and injection effects	Frame-dependent model sensitivity or mismatch	Independent registration, radiometric and geometry checks
Geometry reassignment leaves scores largely unchanged	Weak demonstrated geometry specificity under this stress test	Model adequacy and independent controls
Known heights are recovered with reliable uncertainty coverage	Height inference is supported in the tested regime	Held-out transfer and vehicle-specific evaluation
No useful detection/false-alarm tradeoff exists in the intended regime	Current ranker is unsuitable for the intended decision	Revise or withhold the ranker from that product

These findings need not be mutually exclusive. None turns an unqualified low score into clearance.

The method must ultimately show discriminating performance under independent evidence. The tests are intended to evaluate the existing interpretation, not preserve it.

14.1 Where the campaign results land

Two rows of the table apply at once. Correct geometry improves the fit, so some illumination-dependent structure is being modelled. Geometry reassignment and frame ablations leave most warnings in place, and a changing background with no caster triggers warnings in every trial, so specificity for discrete casters is not demonstrated. With the literature in section 8, the most economical reading is real, illumination-dependent relief broader than the caster template, together with an uncalibrated noise scale. That reading is a hypothesis for tests T12 to T14, not a conclusion.

15 What would retire the method

The likelihood ranker should be withheld from operational hazard decisions if independent evaluation shows that it cannot achieve a useful false-alarm and missed-hazard tradeoff in its intended regime, or if apparent height inference remains unidentifiable or badly miscalibrated. Continued exploration as a research diagnostic is a separate decision.

Geometry stress failure or strong frame dependence would justify investigation and potentially withholding a specific configuration, but neither alone proves that no shadow information is present. Any repair must demonstrate improvement on data not used to choose that repair.

Disagreement with the kinematics voter is a valuable diagnostic, not an automatic win for the voter. The two analyses have different sensitivities and failure modes. Compare both against known or annotated scenes with matched support; a negative voter result can be a missed detection as well as evidence against the ranker.

Independent annotation, realistic injection and false-alarm controls should be prepared before the next scientific performance claim. Their feasibility and cost vary; a new held-out polar ingestion is not equivalent to a cheap saved-array check.

On present evidence the ranker is not ready for operational hazard decisions: under the changing-background control it shows no useful false-alarm tradeoff. Its value as a research diagnostic remains open, because correct geometry measurably improves its fit.

16 Notes on the external feasibility report

The separate report titled "Methodological and Statistical Feasibility of Sub-Pixel Hazard Saturation and Shadow Indexing at the Mons Mouton Lunar Landing Site" motivates considering real high occupancy. Its underlying document and source references must be checked directly before attributing detailed errors or accepting its geological and percolation claims. This text does not independently validate that external report.

For comparison with any proposed explanation, four implementation facts are established here. The production ranker fits a multi-frame intensity stack jointly; it is not a single-image dark-pixel count. The touchdown cell lacks the required common support. Registration sigma is an assumed uncertainty, not an imposed half-pixel translation. The original 10.58-metre warning inheritance arose from painting a cell maximum before buffering; 2.5.5 uses actual root distances.

A predicted coverage percentage from another method cannot be compared with HATI's threshold-exceedance fraction without matching the quantity, scale, mask and detection assumptions. Geological plausibility is a reason to test the physical hypothesis, not a calibration of the existing score.

A literature search through the Consensus service on 24 September 2026 reached a view close to section 8: high occupancy is plausible mainly through small craters and textured regolith rather than boulders. Its citations were checked one by one. Most were confirmed. One paper was dated a year early, and one conference abstract was cited under a later year. Some numerical claims could not be confirmed at source and are not used here, including a global mean rock fraction, a rock survival half-life, a reliable shadow-size range and a distance within which elephant-hide texture occurs.

17 References

Literature cited in sections 8, 11 and 13. Links were checked on 24 September 2026. Conference abstracts, preprints and project pages are marked as such; they have not been through full peer review.

- Amaro et al. (2026). Effect of boulder-size distributions on thermally derived rock abundances on the Moon. *Journal of Geophysical Research: Planets*. <https://doi.org/10.1029/2024JE008769>
- Annex (project page). Artemis III landing site digital terrain modeling via shape-from-shading. <https://andrewannex.com/project/artemissfs/>
- Bandfield et al. (2011). Lunar surface rock abundance and regolith fines temperatures derived from LRO Diviner Radiometer data. *Journal of Geophysical Research* 116, E00H02. <https://doi.org/10.1029/2011JE003866>
- Brown et al. (2025). Grid-mapping of elephant hide textures near the lunar south pole. Lunar and Planetary Science Conference, abstract 2815 (conference abstract). <https://www.hou.usra.edu/meetings/lpsc2025/pdf/2815.pdf>
- Cadogan (2020). Automated precision counting of very small craters at lunar landing sites. *Icarus* 348, 113822.
- Cai and Fa (2020). Meter-scale topographic roughness of the Moon: the effect of small impact craters. *Journal of Geophysical Research: Planets*. <https://doi.org/10.1029/2020JE006429>
- Dagar et al. (2022). Analysis of boulders population around a young crater using very high resolution image of Orbiter High Resolution Camera (OHRC) on board Chandrayaan-2 mission. *Icarus* 386, 115168.
- DEM refinement and validation on the lunar surface using shape-from-shading with Chandrayaan-2 OHRC imagery (2026, preprint). <https://arxiv.org/abs/2604.17436>
- Enhanced topography models for selected lunar south pole regions with shape-from-shading. *Planetary Science Journal*. <https://doi.org/10.3847/PSJ/ae5b70>
- Golombek and Rapp (1997). Size-frequency distributions of rocks on Mars and Earth analog sites: implications for future landed missions. *Journal of Geophysical Research: Planets*. <https://doi.org/10.1029/96JE03319>
- Golombek et al. (2012). Detection and characterization of rocks and rock size-frequency distributions at the final four Mars Science Laboratory landing sites. *Mars* 7, 1. <http://www.marsjournal.org/contents/2012/0001/>
- HATI architecture and implementation review (24 September 2026), implementation 40cf640. Documents/HATI_architecture_and_implementation_review.pdf
- Li and Wu (2018). Analysis of rock abundance on lunar surface from orbital and descent images using automatic rock detection. *Journal of Geophysical Research: Planets* 123, 1061-1088. <https://doi.org/10.1029/2017JE005496>
- Minton et al. (2019). The equilibrium size-frequency distribution of small craters reveals the effects of distal ejecta on lunar landscape morphology. *Icarus*. <https://arxiv.org/abs/1902.07746>
- Powell et al. (2023). High-resolution nighttime temperature and rock abundance mapping of the Moon using the Diviner Lunar Radiometer Experiment with a model for topographic removal. *Journal of Geophysical Research: Planets* 128, e2022JE007532. <https://doi.org/10.1029/2022JE007532>
- Sub-metre lunar DEM generation and validation from Chandrayaan-2 OHRC multi-view imagery using an open-source pipeline (2026, preprint). <https://arxiv.org/abs/2604.01032>

- Xiao and Werner (2015). Size-frequency distribution of crater populations in equilibrium on the Moon. *Journal of Geophysical Research: Planets* 120, 2277-2292.
<https://doi.org/10.1002/2015JE004860>

18 Provenance

Item	Value
Original science run	20260912T072640Z on DESKTOP-FLD4U9E
Original code / version	bbbef4f559df2e25369fc514f2927d78be32060f / 2.5.3
Diagnostic release	66a1cf3a2fd9012b975ecae06a99432f12be3b44 / 2.5.4
Later replay	2.5.5 source hashes recorded in replay run.json
Reference assumptions	registration sigma 0.5 px; noise sigma 0.03
Vehicle limits	configs/landing_illustrative.json; not verified Athena specifications
Source manifest SHA-256	ba7561796c16104ed210d04227ece63fa80c0cd9d292b35c732f4a06310d9dd7
Aligned-stack SHA-256	5a2c06b96eddc4a9f25665c15049405677633eb0a3905f1e2f15f78910db2e39
Frame and bank statistics	verified_run_data.json; original run.json, candidates.csv and coreg_report.csv
Original environment	Python 3.13.15; NumPy 2.5.2; SciPy 1.18.0; Rasterio 1.5.1
Text corrected	19 September 2026
Architecture review	24 September 2026; implementation 40cf64075c331e19d7fb57c4b2b4b3b4de17be66
Campaign archive	athena-watch-01; source revision 8347e6dbfc7532bb59c3311ff888ec3f79efd9a
Literature search	Consensus, 24 September 2026; each cited claim checked at source
Edition 2	24 September 2026

Original-run statistics, diagnostic experiments and later replay results are separate evidence. Proposed tests are not completed experiments. Campaign results are transcribed from the architecture review; nothing was re-run for this document.

Primary external sources used to bound the interpretation:

- LROC NOBILE03 product README: https://pds.lroc.im-ldi.com/data/LRO-L-LROC-5-RDR-V1.0/LROLRC_2001/DATA/SDP/NAC_DTM/NOBILE03/NAC_DTM_NOBILE03_README.TXT . Its confidence codes 10-14 mean successful correlation, not measured local height accuracy. It reports 4 m posting, 3.7 m SOCET precision error and 0.42 m LOLA RMS, which describe different quantities.
- Diviner instrument specifications: <https://www.diviner.ucla.edu/instrument-specs> . The nominal footprint at 50 km altitude is approximately 179 by 307 m; product-specific support and coverage must be checked separately.
- Vanga et al., Rock Abundance on the Lunar Mare on Surfaces of Different Age, 2022: <https://ntrs.nasa.gov/api/citations/20220002292/downloads/VangaetalAccepted.pdf> . This study concerns mare terrains and cannot supply Mons Mouton's rock abundance; it illustrates why older terrain does not automatically mean more exposed rocks.
- Exploring the Mons Malapert and Mons Mouton region, LPSC 2026 abstract 1214: <https://www.hou.usra.edu/meetings/lpsc2026/pdf/1214.pdf> . Its approximately 4.26 Ga value is a geological model age, not a direct measurement of current sub-metre rock density.

WHAT THIS DOCUMENT DOES NOT CLAIM

This report does not establish a subpixel detection at Athena, a measured rock-occupancy fraction, validated height estimates, a probability of mission loss or a landing clearance. It preserves the real-hazard hypothesis while distinguishing observed scores from physical interpretations and identifying the independent tests still required.